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Review Article

Long-term outcomes of liver transplantation versus liver resection for combined hepatocellular–cholangiocarcinoma: An updated systematic review and meta-analysis

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ABSTRACT

Background: Combined hepatocellular cholangiocarcinoma (cHCC-CCA) is a rare and aggressive primary liver cancer. Liver resection (LR) has long been the standard treatment for cHCC-CCA, but it is associated with high recurrence rates and poor long-term prognosis. The significance of liver transplantation (LT) remains controversial. This systematic review and meta-analysis aimed to compare the long-term survival and recurrence outcomes between LT and LR for cHCC-CCA to assess the potential benefit of LT.

Methods: A systematic search of Web of Science, Medline Ovid, Scopus, and Cochrane CENTRAL was conducted using predefined terms related to cHCC-CCA. Records were screened according to PRISMA. Study quality was assessed using the Newcastle-Ottawa Scale (NOS). The primary outcome was 5-year Overall Survival (OS). Recurrence-free survival (RFS) was a secondary outcome. Meta-analysis was performed using random-effects models.

Results: Six retrospective studies met inclusion criteria. NOS scores ranged from 7 to 9. For OS, pooled analyses yielded ORs of 0.67 (95% CI: 0.39–1.15, $p = 0.15$) at 1 year, 0.70 (95% CI: 0.42–1.17, $p = 0.15$) at 3 years, and 0.55 (95% CI: 0.38–0.81, $p = 0.002$) at 5 years. For RFS, pooled analyses yielded ORs of 0.48 (95% CI: 0.30–0.78, $p = 0.003$) at 1 year and 0.40 (95% CI: 0.27–0.59, $p < 0.001$) at 5 years.

Conclusion: LT may be associated with favorable long-term survival and recurrence outcomes compared with LR for cHCC-CCA. These findings suggest that LT could be considered in carefully selected patients.

1. Introduction

Combined hepatocellular cholangiocarcinoma (cHCC-CCA) is a distinct and rare primary liver malignancy characterized by the presence of both hepatocellular carcinoma (HCC) and cholangiocarcinoma (CCA) components [1–3]. Due to its rarity, the clinical guidelines for its management are not as well-established as those for isolated HCC or CCA [3–5]. Because of its aggressive biology, cHCC-CCA has long been associated with poor prognosis [6,7].

Surgical intervention remains the primary treatment strategy [8,9]. Historically, liver resection (LR) has been the standard approach, but it is associated with a high rate of recurrence due to the tendency for intrahepatic metastasis [10]. Additional factors such as lymph node metastasis, vascular invasion, and insufficient resection margins (R1 resection) frequently dictate poor outcomes [11].

Liver transplantation (LT) has been successfully employed for unresectable HCC, offering the potential for complete tumor removal and treatment of underlying liver disease [12]. However, LT was historically

Abbreviations: AFP, alpha-fetoprotein; CA 19-9, carbohydrate antigen 19-9; CCA, cholangiocarcinoma; cHCC-CCA, combined hepatocellular–cholangiocarcinoma; CI, confidence interval; HCC, hepatocellular carcinoma; ICC, intrahepatic cholangiocarcinoma; IQR, interquartile range; LT, liver transplantation; LR, liver resection; MVI, microvascular invasion; NOS, Newcastle-Ottawa quality assessment scale; OR, odds ratio; OS, overall survival; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PROSPERO, the International Prospective Register of Systematic Reviews; pCCA, perihilar cholangiocarcinoma; PSM, Propensity Score Matching; RFS, recurrence-free survival.

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considered contraindicated for CCA due to poor post-transplant outcomes and high recurrence rates, particularly for intrahepatic and perihilar subtypes [13,14]. This paradigm shifted following the successful application of LT for perihilar CCA (pCCA), particularly using the Mayo Clinic protocol, which combines strict patient selection criteria with neoadjuvant chemoradiotherapy [15,16]. This protocol has demonstrated markedly improved survival for pCCA, leading to its adoption as standard of care in experienced centers. This success has fueled optimism that a similar rigorous protocol applied to intrahepatic cholangiocarcinoma (ICC) could overcome the limitations of LR and yield superior long-term survival [17,18]. Given the mixed nature of cHCC-CCA, the efficacy of LT in this patient group has been the subject of ongoing research [19,20].

In recent years, several retrospective studies comparing the outcomes of LT and LR for cHCC-CCA have been published [19,20]. However, the results remain inconsistent, necessitating a comprehensive integration of evidence, particularly concerning long-term prognosis. The objective of this systematic review and meta-analysis is to synthesize the current evidence from comparative studies of LT versus LR for cHCC-CCA and provide a detailed statistical comparison of long-term overall survival (OS) and recurrence-free survival (RFS) outcomes.

2. Materials and methods

2.1. Search strategy and study selection

This systematic review and meta-analysis were conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [21]. The study protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO; registration number CRD420251180496). We conducted a systematic search of the Web of Science, Medline Ovid, Scopus, and Cochrane CENTRAL databases for relevant articles published up to October 31st, 2025. The search utilized a combination of keywords related to cHCC-CCA, LT, LR, survival, and recurrence (Supplemental Table 1).

We included all studies that reported comparing long-term outcomes in LT and LR for cHCC-CCA. If multiple studies were published by the same institution or using the same database, we selected the most recent article. Records without abstracts, case reports, letters, editorials, non-comparative studies, conference abstracts, and reviews were excluded from this analysis. Duplicate records were removed, and two investigators determined eligible papers for further analysis by screening of titles and abstracts independently. Full-text articles were subsequently retrieved and examined independently by two investigators to ensure they met the inclusion criteria and included at least one outcome variable. Then, we extracted the following information: year and country of study publication, study design, and postoperative long-term outcomes.

2.2. Quality assessment

The methodological quality of each study was evaluated based on the Newcastle-Ottawa quality assessment scale (NOS) for cohort studies [22]. The NOS evaluates studies based on three domains: selection of study groups, comparability of groups, and ascertainment of the outcome. Studies with a total score of 6 or higher were considered high quality (Supplemental Table 2).

2.3. Outcome variables

The primary outcome was 5-year OS in LT and LR. The secondary outcomes included 1-year and 3-year OS, and 1-year and 5-year RFS.

2.4. Statistical analysis

Values are presented as proportions for categorical data and as averages with standardized differences or medians with interquartile ranges for continuous variables. All reported p values were two-tailed, with an alpha level of 0.05 considered statistically significant.

Meta-analysis was conducted using RevMan (Version 5.4) to account for between-study variability. Effect sizes were pooled as odds ratios (OR) with corresponding 95% confidence intervals (CI) for each outcome (e.g., 5-year OS, 5-year RFS). Statistical heterogeneity among studies was assessed using the I^2 statistic and Cochran's Q test. An I^2 value of $<25\%$ was considered low heterogeneity, 25–50% moderate, and $>50\%$ substantial heterogeneity. A random-effects model was used when substantial heterogeneity was present. Sensitivity analyses were conducted by sequentially excluding individual studies (leave-one-out analysis) to evaluate the robustness of the pooled estimates. Additional analyses were performed to assess the impact of study-level characteristics where applicable.

3. Results

3.1. Characteristics and quality of included studies

The initial database search yielded 603 records, with 564 remaining after the removal of duplicates. The screening process, summarized by the PRISMA 2020 flow diagram (Fig. 1), excluded 542 records for reasons including: not an English paper (n = 28), not original articles (n = 202), other diseases (n = 35), or other topics (n = 277). Full-text articles were assessed for eligibility, and 16 were excluded primarily because the study included other diseases, necessary data were not available, or data were overlapped. Ultimately, six studies were included in the qualitative synthesis and meta-analysis [19,20,23–26].

The characteristics of the included studies are summarized in Table 1. All six included studies were retrospective in design, spanning years from 2013 to 2023. The six included studies collectively analyzed outcomes for patients undergoing LT (n = 215) and LR (n = 410) for cHCC-CCA. Most studies were single-center retrospective studies, although three were nationwide or multi-center. The studies varied in geographic origin, including the United States, South Korea, Taiwan, and China. The indication for LT was often an incidental finding of cHCC-CCA in the explanted liver, though this was only specifically reported in one study. Of the LT patients whose tumor stage was known, 92 (94.8%) had a pathological stage of I or II, as did 243 (91.4%) of the LR patients (p = 0.37). Additionally, detailed tumor characteristics are presented in the Supplemental Table 3. The mean tumor size was 2.19-2.57 in the LT group and 2.7-6.26 in the LR group, indicating a tendency toward larger sizes in the LR group. The proportion of patients with solitary tumors was 0-87.5% in the LT group and 70.6-100% in the LR group. The proportion of microvascular invasion (MVI) differed between studies, ranging from 18.8% to 62.5% in the LT group and from 25.0% to 76.5% in the LR group.

The methodological quality of the included studies is summarized in Supplemental Table 2. The total NOS score for the included studies ranged from 7.5 to 9 out of 9, indicating overall good to high quality. The study by Mi et al. received the highest score 9. Notably, one study utilized statistical techniques like Propensity Score Matching (PSM) to minimize the impact of confounding factors.

3.2. Meta-analysis: comparison of survival rates

The starting point for survival measurement differed across studies, with one study reporting outcomes from diagnosis, one from surgery, and the remaining studies not specifying the time origin. The comparison of OS rates between the LT and LR groups is summarized in Table 2 and Fig. 2 (a, b, and c). Heterogeneity was low to moderate across analyses. For 1-year OS, the pooled OR was 0.67 (95% CI: 0.39–1.15, p =

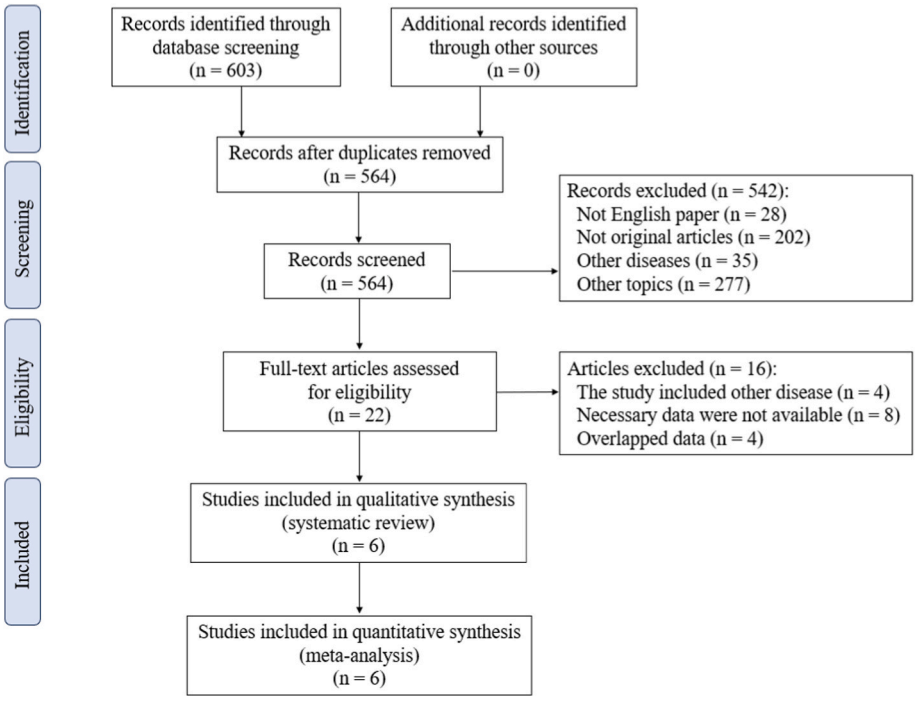


Fig. 1. PRISMA 2020 flow diagram of articles included in the present review.

Table 1
Summary of the studies included in the present review.

			Study	Study	Indication of	Number of patients		Male (%)		Age		Tumor Stage I/II/III/IV	
Authors	Year	Country	setting	design	LT for cHCC-CCA	LT	LR	LT	LR	LT	LR	LT	LR
Groeschl et al.	2013	USA	nationwide	retrospective	not described	19	35	NA	NA	NA	NA	NA	NA
Song et al.	2013	South Korea	single center	retrospective	not described	8	68	8 (100)	51 (75)	53.8 ± 8.2	55.7 ± 1.1	0/8/0/0	15/53/0/0
Wu et al.	2016	Taiwan	single center	retrospective	not described	8	24	NA	NA	NA	NA	8 (I+II)/ 0/0	24 (I+II)/ 0/0
Jung et al.	2017	South Korea	single center	retrospective, PSM	incidental finding of explanted liver	32	100	26 (81.3)	75 (75.0)	53.4 ± 6.7	53.9 ± 10.6	23/9/0/0	75/25/0/0
Dageforde et al.	2021	USA	multi-center	retrospective	not described	99	109	75 (75.8)	71 (65.1)	58.5 ± 9.5	61.6 ± 12.8	NA	NA
Mi et al.	2023	China	nationwide	retrospective	not described	49	74	39 (79.6)	46 (62.2)	NA	NA	44(I+II)/5 (III+IV)	51(I+II)/ 23(III+IV)

cHCC-CCA, combined hepatocellular cholangiocarcinoma; DFS, disease-free survival; LR, liver resection; LT, liver transplantation; NA, not applicable; PSM, propensity score matching.

Table 2
Comparison of the long-term survival rate between LT and LR in the included studies.

			Number of patients		Time origin of	1-year OS (%)		3-year OS (%)		5-year OS (%)		1-year RFS (%)		5-year RFS (%)	
Authors	Year	Country	LT	LR	survival	LT	LR	LT	LR	LT	LR	LT	LR	LT	LR
Groeschl et al.	2013	USA	19	35	Not reported	89.0	71.0	48.0	46.0	NA	NA	NA	NA	NA	NA
Song et al.	2013	South Korea	8	68	Surgery	NA	NA	NA	NA	50.0	42.1	NA	NA	NA	NA
Wu et al.	2016	Taiwan	8	24	Not reported	NA	NA	75.0	76.0	NA	NA	NA	NA	NA	NA
Jung et al.	2017	South Korea	32	100	Not reported	84.4	85.0	73.1	77.5	65.8	63.0	84.4	77.0	67.8	43.7
Dageforde et al.	2021	USA	99	109	Not reported	83.3	81.4	66.7	56.7	58.6	43.8	79.0	65.2	65.2	47.1
Mi et al.	2023	China	49	74	Diagnosis	86.2	70.3	72.4	44.4	60.3	34.5	90.0	74.8	69.3	42.9

DFS, disease-free survival; LR, liver resection; LT, liver transplantation; NA, not applicable; OS, overall survival; RFS, recurrence-free survival.

0.15; $I^2 = 17\%$). Individual study 1-year OS rates for LT ranged from 83.3% to 89.0%, and for LR, from 70.3% to 85.0%. For 3-year OS, the pooled OR was 0.70 (95% CI: 0.42–1.17, $p = 0.15$; $I^2 = 38\%$). For 5-year OS, the pooled OR was 0.55 (95% CI: 0.38–0.81, $p = 0.002$; $I^2 = 0\%$). Individual study 5-year OS rates for LT ranged from 50.0% to 65.8%, and for LR, from 34.5% to 63.0%.

The comparison of RFS rates between the LT and LR groups is summarized in Table 2 and Fig. 2(d and e). For 1-year RFS, the pooled OR was 0.48 (95% CI: 0.30–0.78, $p = 0.003$; $I^2 = 0\%$). For 5-year RFS, the pooled OR was 0.40 (95% CI: 0.27–0.59, $p < 0.001$; $I^2 = 0\%$). Individual

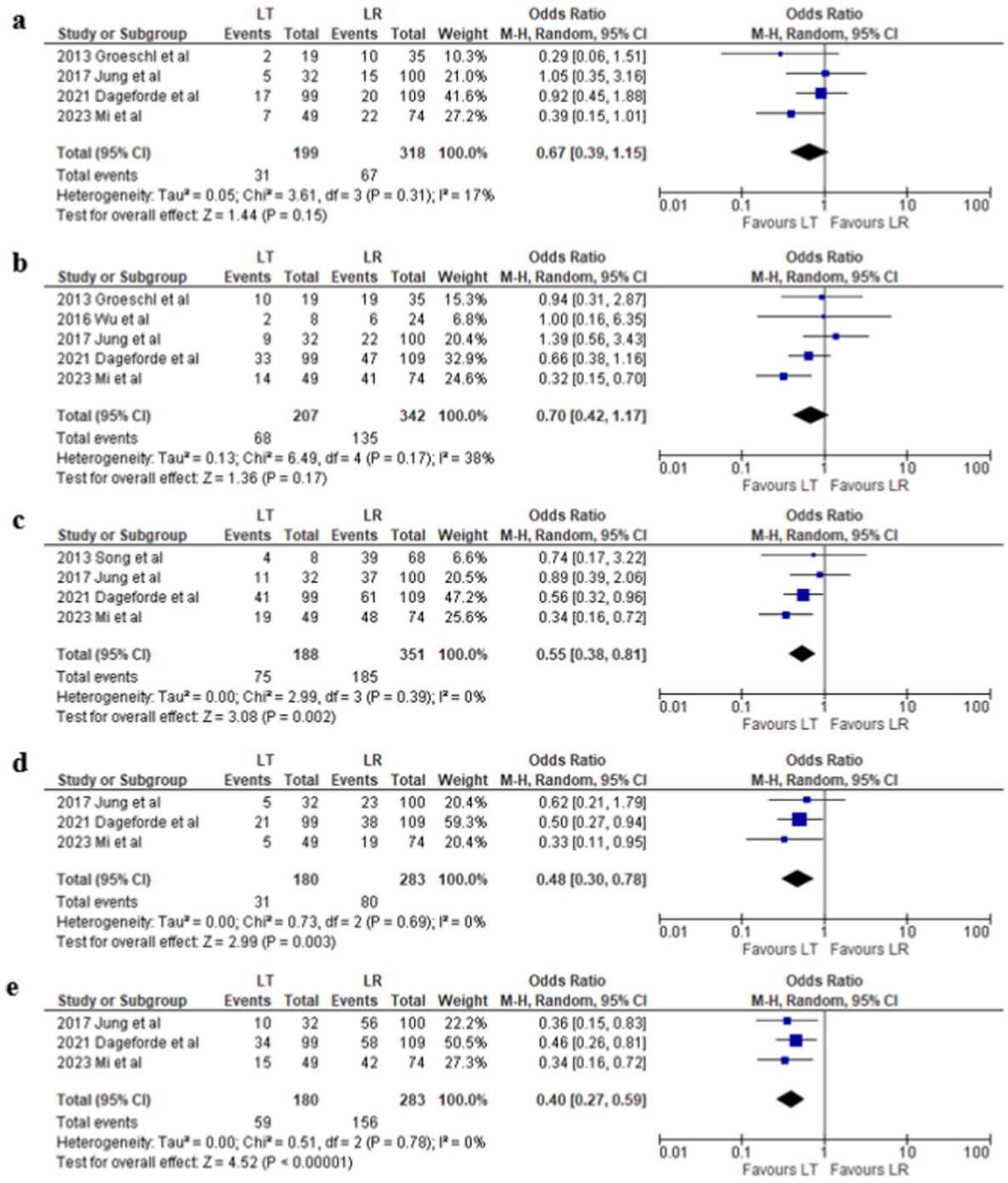


Fig. 2. Forest plots comparing survival rate between LT and LR. (a) 1-year OS, (b) 3-year OS, (c) 5-year OS, (d) 1-year RFS, (e) 5-year RFS. LR, liver resection; LT, liver transplantation; OS, overall survival; RFS, recurrence-free survival.

study 5-year RFS rates for LT ranged from 65.2% to 69.3%, and for LR, from 42.9% to 47.1%.

Sensitivity analyses were performed using a leave-one-out approach to assess the robustness of the pooled estimates. For OS and RFS outcomes, the overall direction of effect remained consistent after sequential exclusion of individual studies. However, for 3-year and 5-year OS, exclusion of one study resulted in changes in statistical significance, indicating some degree of influence from individual studies. In contrast, 1-year OS and all RFS analyses remained robust, with no material changes in effect estimates.

4. Discussion

This updated systematic review and meta-analysis of six retrospective studies demonstrates that LT is associated with significantly improved long-term outcomes compared with LR for cHCC-CCA, particularly with respect to 5-year OS and both early and late RFS. Sensitivity analyses demonstrated that 1-year OS and all RFS outcomes were robust to study-level differences, whereas 3-year OS showed some sensitivity to the exclusion of individual studies, indicating potential influence of between-study variability. Importantly, these benefits were observed in cohorts overwhelmingly composed of patients with early-stage tumors, highlighting the critical role of tumor biology and stage

in determining the appropriateness of LT for this historically high-risk malignancy. These results indicate a dramatic improvement in LT outcomes for cHCC-CCA, moving past earlier reports that showed dismal prognosis [1,27].

A key observation across the included studies is that the vast majority of transplanted patients had pathological stage I–II disease [20,28]. There have not been universally adopted criteria for LT. However, most of patients with cHCC-CCA had LT within Milan Criteria, which include a single lesion ≤5 cm or up to three lesions each ≤3 cm, with no macrovascular invasion or extrahepatic disease [19]. In addition, a major pitfall of LR is the high rate of early recurrence, often due to microscopic satellite lesions or occult lymph node metastasis resulting in an R1 margin [10]. The significant improvement in RFS with LT is expected, as LT removes the entire native liver, eliminating occult intrahepatic metastases and the underlying diseased liver parenchyma, which is a major site of recurrence following LR [29–31]. This completeness of tumor removal, which is often challenging to achieve with LR, is the likely primary driver behind the significant superiority observed in the 5-year RFS. The 5-year OS advantage further reinforces that the curative potential of LT outweighs the associated operative risk and the risks of long-term immunosuppression [14,19].

The heterogeneous biological behavior of cHCC-CCA represents a major challenge in treatment selection [1,3,32]. Pathologically, cHCC-CCA encompasses a spectrum ranging from HCC-dominant tumors with focal cholangiocytic differentiation to tumors with aggressive cholangiocarcinoma-dominant features [2,33]. Prior studies have shown that cholangiocarcinoma-dominant histology, MVI, lymphatic spread, and poor differentiation are strongly associated with early recurrence and poor survival after LR [33, 34]. Unfortunately, these adverse features are often underappreciated preoperatively due to sampling limitations and diagnostic ambiguity.

Tumor markers further reflect this biological heterogeneity. Elevation of AFP, CA19-9, or both has been associated with more aggressive tumor behavior and inferior outcomes [35,36]. Notably, dual-marker elevation (AFP and CA19-9) may represent a distinct high-risk biological phenotype, potentially mirroring biphenotypic tumor differentiation and increased metastatic potential [35]. In this context, LT may partially mitigate adverse biology by removing the entire liver, including microscopic intrahepatic disease that would otherwise contribute to early recurrence after LR [19,20]. However, the benefit of LT is unlikely to be uniform across all biological subtypes. Patients with aggressive pathological features or markedly elevated CA19-9 may remain at high risk for post-transplant recurrence, emphasizing the need for biology-driven selection criteria, rather than reliance on size and number alone [19,37]. In fact, cHCC-CCA is a biologically diverse tumor, and its outcome likely depends on the dominant histologic features, vascular invasion, nodal status, and tumor marker profile.

Recent literature increasingly emphasizes that oncological outcomes after surgical treatment for primary liver malignancies are largely determined by tumor biology rather than the type of surgical procedure alone [38]. In particular, early recurrence—typically defined as recurrence within the first 1–2 years after surgery—has been recognized as a surrogate marker of aggressive tumor biology, reflecting the presence of microscopic metastases or biologically invasive disease at the time of treatment [39]. Several studies have demonstrated that early recurrence is strongly associated with poor long-term survival following hepatic resection for primary liver cancers [30,31]. In the context of cHCC-CCA, the high incidence of early intrahepatic recurrence reported after LR likely reflects the underlying aggressive biology of this tumor. Therefore, improved RFS observed after LT in the present analysis may partially result from the complete replacement of the diseased liver and elimination of occult intrahepatic tumor dissemination. Interestingly, Jung et al. reported that 71.4% of patients in the LR group developed intrahepatic recurrence, whereas only 16.7% of patients in the LT group experienced intrahepatic recurrence [24]. These observations further support the concept that careful biological selection, rather than surgical modality alone, is critical for optimizing oncological outcomes in patients with cHCC-CCA.

Unlike pCCA, where the Mayo Clinic protocol established neoadjuvant chemoradiotherapy as a cornerstone of successful LT, there is currently no standardized perioperative systemic therapy strategy for cHCC-CCA or ICC [40,41]. In the studies included in this meta-analysis, the use of neoadjuvant or adjuvant chemotherapy was inconsistent and poorly reported, precluding meaningful subgroup analysis. Nevertheless, given the recognized cholangiocarcinoma-like behavior of many cHCC-CCA tumors, systemic therapy may play an increasingly important role. Platinum-based chemotherapy, gemcitabine-based regimens, and emerging targeted therapies could potentially improve post-LT disease control, particularly for tumors with adverse pathological features [42,43]. From a biological standpoint, LT alone may be insufficient for tumors with aggressive molecular or pathological signatures, and integration of systemic therapy may be necessary to replicate the success seen in pCCA [19]. However, LT success in pCCA is closely linked to strict selection criteria and neoadjuvant therapy protocols. In contrast, ICC remains a controversial transplant indication with limited support, even for highly selected patients. For this reason, cHCC-CCA should not be implicitly considered analogous to pCCA due to the biological similarity between cHCC-CCA and ICC. In addition, unlike pCCA, many cases are diagnosed incidentally on explant pathology, and some cases do not have a preoperative biopsy. Even when there is a biopsy, it can be difficult to make a preoperative diagnosis. These factors can make preoperative diagnosis difficult.

The findings of this meta-analysis challenge the historical dogma that cHCC-CCA represents an absolute contraindication to LT. Instead, they suggest that early-stage cHCC-CCA with favorable biological characteristics behaves more like HCC than CCA in the transplant setting, achieving survival outcomes comparable to accepted LT indications. Given the biological heterogeneity of cHCC-CCA, transplant candidacy may depend on the dominant tumor phenotype and biomarker profile. Tumors with HCC-like features and lower CA19-9 levels appear to have more favorable outcomes after LT, whereas CCA-dominant or CA19-9–elevated tumors may be associated with higher recurrence risk [33,44,45]. Importantly, these results support a paradigm shift in which LT should not be reflexively excluded. However, the apparent advantage of LT may be partly due to patient selection rather than the effect of the treatment itself. In fact, cHCC-CCA is a distinct entity with variable biology that overlaps with both HCC- and CCA-like behavior. Therefore, future studies should clarify which tumor biological factors contribute to the criteria for choosing a surgical treatment plan.

This study is limited by the retrospective nature of the included analyses, heterogeneity in selection criteria, and incomplete reporting of pathological subtypes, tumor markers, and chemotherapy use. Additionally, most LT cases were incidental diagnoses, limiting the ability to prospectively apply selection criteria. These findings, derived from retrospective analyses, suggest that LT may be associated with favorable outcomes. However, prospective studies are needed to confirm its applicability in clinical practice. Another important limitation of this study is the variability in the definition of survival outcomes across included studies. Differences in the time origin of survival measurement (e.g., diagnosis versus surgery) may introduce bias and limit the comparability of pooled estimates. Therefore, the results should be interpreted with caution.

Despite these limitations, the consistency of long-term RFS benefit across studies strengthens the validity of the conclusions. In addition, assessment of publication bias was limited because fewer than 10 studies were included in the meta-analysis.

Future research should focus on: 1. defining biological selection criteria incorporating pathology, tumor markers, and molecular profiling; 2. establishing standardized peri-transplant chemotherapy protocols; 3. developing prospective multicenter registries to validate LT indications for cHCC-CCA.

5. Conclusions

LT was potentially associated with favorable long-term survival and recurrence outcomes compared with LR for cHCC-CCA in the included studies. These findings suggest that LT may benefit a highly selected subgroup of patients with favorable disease biology and limited tumor burden. Further prospective studies are needed to confirm these observations.

Availability of data and materials

All data supporting the conclusions of this article are included within the
Published article.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

CRediT author statement

Jiro Kimura: Conceptualization, Methodology, Software, Writing-Original draft preparation, Writing- Reviewing and Editing, Visualization, Investigation, Validation, Ayham Asassfeh: Investigation, Data curation, Matthew Cooper and Kondragunta Rajendra Prasad: Supervision.

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Conflict of interest

The authors declare no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejso.2026.111846.

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